WHITE PAPER · HWACHEON MEGA 100 × MILITARY & DEFENSE

Hwacheon Mega 100 for Military & Defense

Heavy-duty CNC lathe with industry-leading 10" thru-hole for tubing, collars, subs, and long-shaft work. This paper covers what the machine does, the military & defense parts we produce with it, the alloys we run, the tolerances and finishes military & defense buyers expect, and the documentation package.

16" OD × 100" L × 10" thru-hole ENVELOPE	**±0.0005" routine on critical features** TOLERANCE	**1994** SINCE
Same-week / rig-down direct RESPONSE		



Texaco Thunder · 16" OD × 100" L × 10" thru-hole · heavy-duty CNC lathe with 10" thru-hole

PUBLISHED
January 15, 2026

LAST UPDATED
2026-01-15

FORMAT
Web + PDF

READ TIME
~6 min

Executive summary

Machine: Hwacheon Mega 100 (shop nickname: *Texaco Thunder*). heavy-duty CNC lathe with 10" thru-hole at B&R Productions, running from our New Waverly, Texas shop.

Application: Heavy-duty CNC lathe with industry-leading 10" thru-hole for tubing, collars, subs, and long-shaft work for Military & Defense customers — Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

Envelope: 16" OD × 100" L × 10" thru-hole. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

The 10-inch through-hole is where Texaco Thunder earns its keep. Long tubing stock, downhole tool bodies with internal passages, coiled-tubing tool bodies, large valve stems with bore access from both ends — parts that other shops outsource because their machines can't accommodate the stock. This machine turns outsourced-work into in-house work.

Why Texaco Thunder for Defense work

Defense tubular work with real bore requirements — long tubular housings, through-bored bodies, components needing access from both ends — runs into the same spindle-bore ceiling as oilfield work. A 10" through-hole at 16" OD × 100" L is what removes it.

What this looks like on a real part

A long tubular housing that must stay concentric end to end is the argument for feeding stock through the spindle. Cut it short and flip it and you have introduced a re-indication into the exact requirement the drawing cares about most. Fed through, both ends work off the original datum. On sustainment parts this matters doubly, because the replacement has to mate with existing hardware that was never intended to accommodate someone else's tolerance stack.

> **When this is the wrong machine**
>
> Non-controlled work only. 400/750 RPM makes this a slow, deliberate machine — small or speed-sensitive defense parts belong on Bearkat Blur.

Full specifications

Model	Hwacheon Mega 100 ("Texaco Thunder")
Type	Heavy-duty flatbed horizontal CNC lathe — Mega-100 platform
Max turning capacity	16" OD × 100" L (B&R working spec)
Platform max cutting Ø	27.5" (700 mm) on the Mega-100
Chuck swing over bed	42.5" (1,080 mm)
Distance between centers	10 ft (3,000 mm) B&R config; platform available to 20 ft (6,000 mm)
Spindle bore	10" (255 mm) B&R config; platform range 6" – 12.6"
Spindle speed	400 / 750 RPM (two-range)
Spindle motor	25 HP
Turret	4 or 8-station heavy-duty
Machine weight	~35,000 lb (16,000 kg)
Purpose	Long shafts, main-shaft work, large pipe / tubing components
Tolerance capability	±0.0005" routine on critical features
Materials	Full oilfield alloy range — Inconel, Duplex 2205/2507, 17-4 PH, 4140, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Military & Defense

Typical buyer: Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

What's hard about military & defense work: Sensitive prints, program-specific quality plans, tight-tolerance structural components, exotic-alloy selection driven by service environment, and legacy-part sustainment where the OEM no longer supports the design and the prime's supply chain has moved on.

Typical Military & Defense parts we produce on Texaco Thunder

- Long hydraulic cylinders for defense platforms
- Long structural components with internal passages
- Weapon barrel and tube components
- Sensor tube and probe long-form components
- Actuator cylinder bores with 10" thru-hole

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Military & Defense

Standard range: 17-4 PH · 15-5 PH · 13-8 Mo · 4140 pre-hard · 4340 · Titanium Grade 5 · Inconel 718 · Aluminum 7075 · High-strength steels · Customer-specific alloys on request.

Defense parts routinely spec 4140 pre-hard, 4340, and precipitation-hardening stainless — condition-specific machining knowledge matters, same as aerospace. Titanium is common on weight-critical structural components. Legacy sustainment often specifies materials nobody sees anywhere else; we work with the spec sheet you provide and validate machining strategy before we cut.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, military & defense or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Military & Defense

±0.0005" routine on critical features. Tighter with proper setup and fixture. GD&T-driven tolerancing supported. CMM verification on first articles; customer-plan-driven measurement on production runs. Documentation the level a program office expects.

Documentation and quality workflow

Sensitive prints handled under NDA with document control — prints stay in-shop and aren't sent to outside vendors. First-article documentation standard. Material traceability by heat and lot on every job. Certificates of conformance with delivery. One thing we state plainly up front: B&R is not ITAR-registered with the U.S. State Department. If your print is ITAR-controlled, we are not the right shop and we'll tell you that on the first call. Non-

controlled defense-adjacent work — commercial-spec components, sustainment parts, tooling, and fixtures — we quote all day.

Response time and emergency work

Program-driven schedules honored. Emergency component replacement supported for sustainment programs with established relationships. Legacy-part reverse-engineering supported for out-of-production sustainment.

How we compare

Compared to a large defense-prime captive shop: we're faster for one-offs, small runs, and legacy sustainment where the prime's supply chain has moved on. Compared to a lowest-bidder commodity shop: we understand the alloys, we document properly, and we don't guess at a condition-specific machining plan. Compared to an ITAR-registered shop: for controlled prints they win, because that's a State Department registration and we don't hold it. Know which bucket your part is in before you shop it — it saves everyone a wasted quote cycle.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Mega 100 platform and Military & Defense work. Also indexed as FAQ schema for AI answer engines.

What's special about the Mega 100's 10" thru-hole?

It accommodates long bar stock, tubing, and hollow parts that most CNC lathes can't clear. Downhole tool bodies with internal passages, coiled-tubing tool bodies, and long valve stems with bore access from both ends — parts most job shops outsource because their machines can't take the stock.

How long a part can the Mega 100 turn?

B&R's stated envelope is 100" (10 ft) between centers. The Mega-100 platform is available up to 6,000 mm (~20 ft) between centers on longer configurations.

What's the chuck swing and max cutting diameter?

Chuck swing over bed 42.5" (1,080 mm); max cutting Ø on the platform 27.5" (700 mm). Two-range spindle at 400/750 RPM with a 25 HP main motor.

Do you handle ITAR-controlled defense work?

No — and we'd rather be the shop that tells you that up front. ITAR compliance is a registration with the U.S. State Department's DDTC, not a set of practices a shop can simply adopt, and B&R does not hold it. If your print is ITAR-controlled, you need a registered facility. What we do handle: non-controlled defense-adjacent work — commercial-spec components, legacy sustainment parts, tooling, fixtures, and reverse-engineered replacements for programs the OEM walked away from. Sensitive prints stay in-shop under NDA with document control, and first-article documentation and material traceability come standard.

What kind of defense-adjacent parts is this shop suited to?

Structural and mounting hardware, housings and enclosures, hydraulic components, ground-vehicle and equipment parts, tooling and fixtures, and reverse-engineered legacy parts for platforms the OEM no longer supports — all on non-export-controlled work. That's a description of what the equipment and the alloy experience suit, not a claim about specific programs. If you want to know whether we've run something like your part, ask and we'll tell you straight.

What alloys are typical for defense work at B&R?

17-4 PH, 15-5 PH, 4140 pre-hard, 4340, Titanium Grade 5, Inconel 718, aluminum 7075, high-strength steels. Customer-specific alloys on request.

Can you reverse-engineer legacy defense parts?

Yes — reverse-engineering from a worn sample is core job-shop work and it's what a shop like this is for. Send the sample and any documentation that survives; we'll measure it, verify the alloy, and produce the replacement with full material traceability.

What tolerances do you hold on defense components?

±0.0005" routine on critical features. Tighter with the right fixture. Every critical dimension CMM-verified. Documentation to customer spec.

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Ready to talk about your Military & Defense work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.